

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604D - WATER POWER ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Firm power in a hydroelectric plant refers to:
 - A) Power available during peak hours only
 - B) Power that can be generated under minimum flow conditions
 - C) Power generated during floods
 - D) Installed capacity of the plant
 - 2 Load duration curve is obtained by:
 - A) Arranging loads in descending order of magnitude
 - B) Plotting load against time sequence
 - C) Plotting energy consumption against time
 - D) Plotting power factor against load
 - 3 Run-of-river hydro power schemes are characterized by:
 - A) Large storage reservoirs
 - B) Underground power houses only
 - C) Pumped storage operation
 - D) Negligible storage and dependence on river flow
 - 4 Trash racks are provided at intake structures mainly to:
 - A) Reduce velocity of flow
 - B) Prevent entry of debris into conduits
 - C) Increase head
 - D) Reduce sediment deposition
 - 5 Penstocks are designed as economic sections to:
 - A) Minimize head loss and total cost
 - B) Increase flow velocity
 - C) Reduce pressure
 - D) Improve aesthetics

- 6 The primary function of a surge tank is to:
 - A) Store water for power generation
 - B) Reduce water hammer and pressure fluctuations
 - C) Increase discharge
 - D) Remove air from water
- 7 The main advantage of an underground power station is:
 - A) Lower construction cost
 - B) Better protection and reduced environmental impact
 - C) Easier access for maintenance
 - D) Requirement of less excavation
- 8 The sub-structure of a power house mainly supports:
 - A) Transmission lines
 - B) Control room only
 - C) Turbines, generators, and associated equipment
 - D) Intake structure
- 9 Cavitation in turbines occurs when:
 - A) Local pressure falls below vapor pressure
 - B) Pressure exceeds atmospheric pressure
 - C) Velocity is very low
 - D) Tail race level is high
- 10 Draft tubes are provided mainly to:
 - A) Increase turbine speed
 - B) Control flow direction only
 - C) Reduce discharge
 - D) Recover kinetic energy at turbine outlet
- 11 A major limitation of tidal power stations is:
 - A) High efficiency
 - B) Predictable energy
 - C) High initial cost and limited suitable sites
 - D) No environmental impact
- 12 Pumped storage plants operate mainly during:
 - A) Flood seasons
 - B) Off-peak and peak load periods
 - C) Night time only
 - D) Continuous operation

Q2. Solve the following.

- A) Explain sources of energy and types of power plants. Discuss in short criteria for choice of type of power generation. 6
- B) Explain load curve and load duration curve. Define load factor, firm power, and secondary power. 6

Q3. Solve the following.

- A) Explain types of intakes used in hydropower projects and hydraulics of intake structures. 6
- B) Explain surge tanks—functions, behavior, location, and types. 6

Q4. Solve Any Two of the following.

- A) Explain general layout of a hydroelectric power station with neat sketch. 6

- B) Explain underground power stations—necessity, types, and advantages. 6
- C) Explain substructure and superstructure components of a powerhouse. 6
- Q5. Solve Any Two of the following.
- A) Explain classification and characteristics of different types of turbines. 6
- B) Explain cavitation in turbines and methods to prevent it. 6
- C) Explain draft tubes—functions, types, and importance in energy recovery. 6
- Q6. Solve Any Two of the following.
- A) Explain pumped storage power plants—purpose and working. 6
- B) Discuss economics and advantages of pumped storage plants. 6
- C) Explain tidal power stations—necessity, classification, and limitations. 6

***** End *****